

Edge-Based Real-Time Face Recognition Entry System

學校系所：中原大學電機工程學系 系級：電機二乙

學號：11228203 11228205 11228249

姓名：林美伶 張維珍 林依辰

Abstract

This study implements a facial recognition-based access control system using OpenCV for face detection and the OpenVINO platform to accelerate recognition inference operations. The system employs a UART communication interface to control the NUC140 microcontroller, simulating the door lock activation mechanism. Experimental results demonstrate that under ideal conditions, the recognition accuracy for registered faces reaches 100%; under single-eye occlusion conditions, the accuracy remains between 90% and 100%. In mask-wearing tests, the average accuracy across 5 subjects was approximately 65%, while in sunglasses-wearing scenarios, the average accuracy decreased to 39.3%. Furthermore, the system provides multiple feedback mechanisms based on recognition results, including LED indicator displays, LCD animation presentations, and buzzer alerts. Experimental validation confirms that this system maintains stability and practicality under various environmental conditions, demonstrating its feasibility for intelligent access control applications.

Introduction

This project uses facial recognition technology to improve the security and convenience of traditional access control systems. Conventional methods like card swiping or password entry are prone to risks such as card loss, theft, or password leakage. By using a laptop with OpenCV for image capture and OpenVINO to accelerate the recognition process, the system performs real-time facial recognition locally without relying on cloud services, ensuring both speed and privacy. The recognition result is then transmitted via UART to the NUC140 microcontroller, which simulates the unlocking of a door. This demonstrates the feasibility and practicality of applying facial recognition in access control systems.

Material and Method

This system integrates hardware components and software tools to implement facial recognition and data transmission functions, simulating a real-world access control system. It is primarily composed of a camera module, a microcontroller, a communication interface, and related development toolkits. The detailed description is as follows:

1. Hardware

1.1 Webcam : Captures facial images for recognition.

1.2 UART Transmission Cable: Transfers data between the computer and microcontroller.

1.3 Microcontroller (NUC140/M0): Processes commands and controls hardware based on recognition results.

2. Software

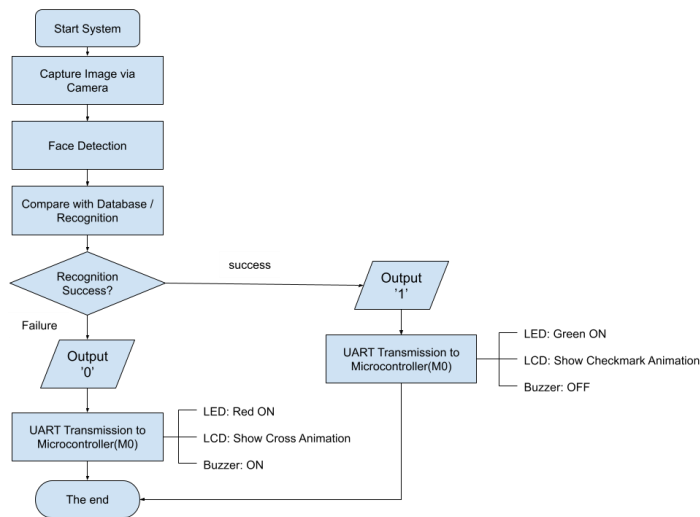
2.1 Programming Languages

- Python : Used for image processing, facial recognition, and serial communication.
- C languages : Programs the microcontroller to handle UART input and control outputs.

2.2 Installed Libraries and Toolkits

- OpenVINO : facial recognition model inference.
- OpenCV : Handles image capture and face detection.
- Pyserial : Enables serial communication between Python and the NUC140.

3.Operation procedure



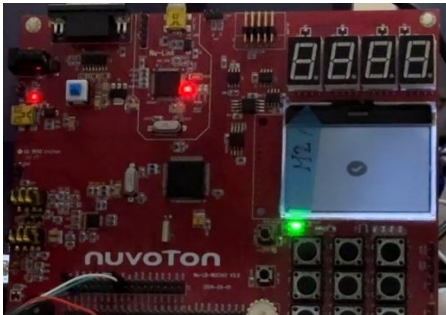
Result

1. The correspondence between experimental inputs and outputs.

This section outlines the system's responses based on different facial recognition inputs. The system behavior varies depending on whether the detected face has been registered in the database.

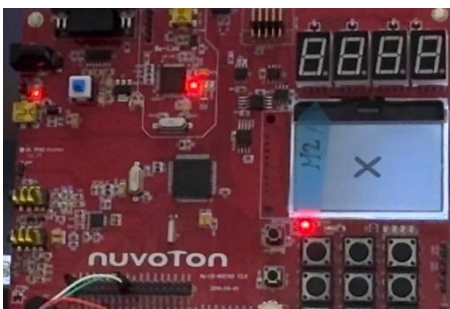
1.1 Input: Registered Face

When a registered face is detected, the system performs the following actions: When the green LED lights up, the LCD screen shows a check mark animation, and the buzzer remains off (OFF).



1.2 Input: Unregistered Face

When an unregistered face is detected, the system performs the following actions : When the red LED lights up, the LCD screen shows a cross mark animation, and the buzzer is activated (ON) to provide a warning.



2. Multiple experiments

In each experiment, six rounds of face recognition tests were conducted, with the system returning a recognition result every three seconds. The participants were divided into two categories: registered users (five individuals) and unregistered or unknown testers (two individuals).

Multiple face recognition tests were conducted to evaluate the system's performance under different user conditions, focusing on both registered and unregistered individuals.

In the first set of experiments, five registered users were tested. Recognition was successful when the full face was visible or when one eye (left or right) was covered, with the system correctly displaying the user's name. However, under low lighting conditions, recognition failed, and the system returned "unknown." For the two unregistered testers, recognition consistently failed even with their full faces visible, and the system correctly identified them as "unknown."

In the second set of experiments, tests were conducted on registered users wearing masks or sunglasses. The system successfully recognized users wearing masks and displayed their names. However, when wearing sunglasses, the results were mixed—some recognitions were successful, while others failed and were marked as "unknown."

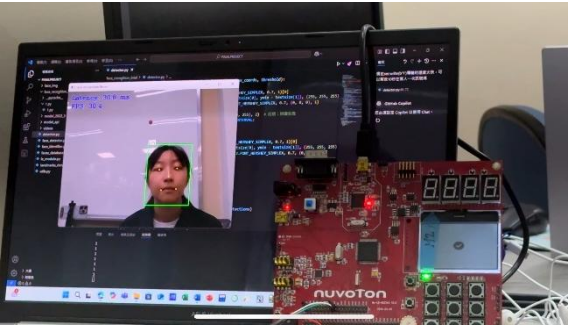
3. Result statistics

3.1 The same input was repeated 6 times

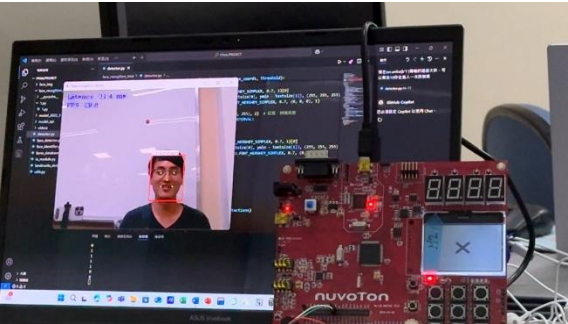
	full face view	cover the left eye	cover the right eye	Low light
karina	100%	100%	100%	70%
lin	100%	80%	90.9%	72.7%
sandy	100%	100%	100%	90%
willy	100%	87.5%	85.7%	70%
tataka	89%	83.3%	85.7%	66.7%
unknown 1	0%			
unknown 2	0%			

3.2 The input conditions were changed and tested 6 times

	Wear sunglasses	Wear mask
karina	0%	85.7%
lin	0%	100%
sandy	33.3%	89%
willy	83.3%	20%
tataka	80%	33.3%
unknown 1		
unknown 2		



Registered user



Unregister individual

Discussion

1. Basic Facial Recognition Performance Analysis

Performance Under Ideal Conditions Under ideal lighting conditions for full-face recognition, the system achieved high accuracy rates for all 5 registered users, with 4 users (karina, lin, sandy, willy) reaching 100% recognition accuracy and tataka achieving 89% accuracy. These results demonstrate that the system possesses good recognition capabilities under standard environmental conditions.

Security Verification for Unregistered Users For the 2 unregistered users (unknown 1, unknown 2), the system correctly identified them as unauthorized personnel (0% pass rate), demonstrating excellent security protection mechanisms.

2. Recognition Capability Under Partial Occlusion Conditions
Single-Eye Occlusion Test Results

- Left eye occlusion: All users maintained 100% accuracy except lin (80%) and tataka (83.3%)
- Right eye occlusion: Accuracy ranged from 85.7%-100%, with overall performance slightly lower than left eye occlusion

These results may reflect:

- Differences in the facial recognition algorithm's sensitivity to left and right facial features
- Individual facial feature distributions may affect recognition effectiveness during single-eye occlusion
- The system demonstrates good fault tolerance, maintaining high recognition rates even with partial feature loss

3. Environmental Factor Impact Analysis

Impact of Lighting Conditions Under low-light environments, recognition accuracy significantly decreased to the 66.7%-90% range, averaging approximately 75.6%. This phenomenon meets expectations because:

- Insufficient lighting reduces image quality, affecting feature extraction precision
- Shadows and contrast changes may interfere with facial detection algorithms

4. Challenges of Occlusion on Recognition Performance

Sunglasses Testing results align with expected settings, as the system is designed to be unable to recognize faces wearing sunglasses to ensure security. Most users showed extremely low accuracy rates (0%-33.3%) in testing, with only willy (83.3%) and tataka (80%) showing higher recognition rates, possibly due to incomplete sunglasses occlusion or accidental system matching. This design prevents security risks from identity disguise through sunglasses and other occlusion methods. Mask Testing showed relatively stable performance with accuracy rates of 20%-100%, with overall impact less than sunglasses. Lin achieved a perfect score, indicating that upper facial features are sufficient to support effective recognition.

5. System Stability and Practicality Assessment

Individual Difference Analysis Significant differences exist in recognition stability among different users under various conditions:

- High stability: karina and sandy performed stably under most conditions
- Moderate stability: lin and willy showed significant fluctuations under specific conditions

- Relatively unstable: tataka's accuracy was lower than other users under all test conditions

Analysis of Recognition Stability Differences:

- Low accuracy for tataka: Primarily due to training photos taken at overly unusual angles, making it difficult for the system to generalize to normal-angle faces
- High fluctuation for lin and willy: Due to insufficient angle variation in training photos, lacking diverse angular data

Improvement Recommendations: We recommend using data augmentation tools such as Roboflow, which can increase training photo angle diversification through automatic angle transformation, rotation, scaling, and other functions. This enhances the system's adaptability to faces at different angles and improves overall recognition accuracy stability.

Conclusion

1. System Achievements and Functional Overview

This project successfully developed an intelligent access control system based on facial recognition, integrating OpenCV for image capture and OpenVINO for deep learning inference to achieve real-time face identification. The system transmits recognition results via UART communication protocol to the NUC140 microcontroller, completing the simulated access control process.

The control unit features a fully functional user interface, including an LCD for status display, LED indicators, and a buzzer for auditory alerts, providing multi-sensory feedback mechanisms.

Experimental results demonstrated a high recognition accuracy of 89% to 100% for registered users under ideal conditions, and 0% false acceptance rate for unregistered individuals, indicating strong security performance. The overall system operates stably and reliably, meeting initial design goals and demonstrating both practical applicability and future scalability.

2. System Limitations and Challenges

Although the system has achieved its fundamental functions, several limitations were identified during real-world testing:

- **Individual Variability:** Recognition stability varied significantly between users, primarily due to insufficient diversity in the training images (e.g., angle, lighting), which affected the system's generalization capability.
- **Security Risk:** The system may mistakenly accept static photos as real faces, posing a vulnerability to spoofing attacks.
- **Output Stability:** Fluctuations in recognition results within short time frames could lead to misjudgments, affecting user experience.

3. Future Improvements and Development Directions

Short-Term Improvements:

- **Liveness Detection:** Integrate liveness detection technology to prevent photo-based spoofing and enhance security.
- **Stability Filtering Mechanism:** Implement a filtering strategy where recognition results are only accepted if over 80% consistency is achieved within 3 seconds, reducing false triggers.
- **Training Data Augmentation:** Use tools like Roboflow to expand the dataset with varied angles and lighting conditions, enhancing model generalization.

Long-Term Development:

- **Multimodal Technology Integration:** Combine with infrared imaging or 3D structured light to improve recognition stability across various environmental conditions.

Personalized Optimization: Fine-tune the system based on individual facial features to further improve overall performance.

References

Face recognition : https://github.com/glennford49/model_Zoo
 OpenCV : <https://opencv.org>
 OpenVino : <https://www.intel.com/content/www/us/en/developer/tools/openvino-toolkit/overview.html>